

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-51213139})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z} 2233 + \mathbf{Z} (54 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\operatorname{Cl}(K)$ (as HNF ideals). A character x on $\operatorname{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\operatorname{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 12803285 = 0$, of discriminant -51213139 . Class group invariants $(75 \ 5 \ 5)$, class number 1875; the three stored generator ideals (HNF) are $\begin{pmatrix} 5 & 4 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 843y^3 + (-6s + 3)y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1686y^8 + 3221837y^6 - 1656013233y^4 + 646383262318y^2 + 469252113896331$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right)s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21641357825712 = 2^4 \cdot 3^2 \cdot 29^2 \cdot 178700603$:

$$\begin{pmatrix} 31093904922 & 29383305072 & 24897882398 & 19532123808 & 12355677 & 7402462968 & 1666566853 & 334387905 & 16443131492 & 3241511106 \\ 0 & 6 & 5 & 0 & 3 & 3 & 4 & 4 & 5 & 5 \\ 0 & 0 & 29 & 0 & 0 & 11 & 20 & 0 & 27 & 5 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 549 factors with signed exponents of absolute value up to 21397235916377340125450; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 843y^3 + (-6s + 3)y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1686y^8 + 3221837y^6 - 1656013233y^4 + 646383262318y^2 + 469252113896331$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right)s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21641357825712 = 2^4 \cdot 3^2 \cdot 29^2 \cdot 178700603$:

$$\begin{pmatrix} 31093904922 & 29383305072 & 24897882398 & 19532123808 & 12355677 & 7402462968 & 1666566853 & 334387905 & 16443131492 & 3241511106 \\ 0 & 6 & 5 & 0 & 3 & 3 & 4 & 4 & 5 & 5 \\ 0 & 0 & 29 & 0 & 0 & 11 & 20 & 0 & 27 & 5 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 559 factors with signed exponents of absolute value up to 30873431389286011407878; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 843y^3 + (-6s + 3)y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1686y^8 + 3221837y^6 - 1656013233y^4 + 646383262318y^2 + 469252113896331$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right)s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65931043733836 = 2^2 \cdot 29^2 \cdot 19599002299$:

$$\begin{pmatrix} 1136742133342 & 899122946660 & 1136503199418 & 1022028690315 & 333933737358 & 417891789390 & 905292473326 & 569576446612 & 100004718107 & 10943385171 \\ 0 & 58 & 52 & 27 & 27 & 6 & 13 & 7 & 37 & 46 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 31871505781869996976220; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 3108)y^3 + (-38s - 70036)y^2 + (-427s - 187823)y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 - 6201y^8 - 115242y^7 + 22650424y^6 + 1412022498y^5 \\ + 35490927781y^4 + 467314056852y^3 + 3430948274295y^2 + 13084669 \\ 240100y + 20307723437500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 0 \ -1 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right)s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2973038062500 = 2^2 \cdot 3^2 \cdot 5^6 \cdot 11^4 \cdot 19^2$:

$$\begin{pmatrix} 344850 & 219450 & 143275 & 292770 & 63785 & 284884 & 289272 & 91371 & 80991 & 59544 \\ 0 & 6270 & 3135 & 1430 & 885 & 1 & 2199 & 4294 & 971 & 5079 \\ 0 & 0 & 55 & 0 & 50 & 31 & 54 & 52 & 9 & 21 \\ 0 & 0 & 0 & 5 & 0 & 1 & 4 & 3 & 1 & 1 \\ 0 & 0 & 0 & 0 & 5 & 0 & 3 & 2 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 2161798005; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 3108)y^3 + (-38s - 70036)y^2 + (-427s - 187823)y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 - 6201y^8 - 115242y^7 + 22650424y^6 + 1412022498y^5 \\ + 35490927781y^4 + 467314056852y^3 + 3430948274295y^2 + 13084669 \\ 240100y + 20307723437500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 0 \ -1 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right)s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18403652224655 = 5 \cdot 41 \cdot 2791 \cdot 32165501$:

$$\begin{pmatrix} 18403652224655 & 4325831784230 & 10559134755395 & 4788656683339 & 2023752516102 & 386328830085 & 2895444247320 & 10834469962410 & 12723026731928 & 16815974437789 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 205 factors with signed exponents of absolute value up to 1694470041; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 3108)y^3 + (-38s - 70036)y^2 + (-427s - 187823)y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 - 6201y^8 - 115242y^7 + 22650424y^6 + 1412022498y^5 \\ + 35490927781y^4 + 467314056852y^3 + 3430948274295y^2 + 13084669 \\ 240100y + 20307723437500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 0 \ -1 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right) s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $50090564129296 = 2^4 \cdot 3727 \cdot 14153 \cdot 59351$:

$$\begin{pmatrix} 6261320516162 & 4885507525016 & 335993844314 & 222591634233 & 3384391214741 & 5603543033324 & 1559613696108 & 155628573063 & 4143617540797 & 794667437541 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 4141560257; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 3325)y^3 + (16s - 43464)y^2 + (-77s + 273390)y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 6645y^8 - 73614y^7 + 24576112y^6 + 699498510y^5 + \\ & 1372949137y^4 - 60244085592y^3 + 90567820395y^2 + 414148101700y \\ & + 994631470000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9100531118475 = 3^2 \cdot 5^2 \cdot 19^2 \cdot 2039 \cdot 54949$:

$$\begin{pmatrix} 159658440675 & 43546671824 & 90991179348 & 66415675607 & 114167891859 & 8699385398 & 59416791670 & 71546503190 & 135157323602 & 95609813899 \\ 0 & 19 & 0 & 16 & 3 & 17 & 6 & 11 & 3 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 222 factors with signed exponents of absolute value up to 87432026487069863046; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 3325)y^3 + (16s - 43464)y^2 + (-77s + 273390)y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 6645y^8 - 73614y^7 + 24576112y^6 + 699498510y^5 + \\ & 1372949137y^4 - 60244085592y^3 + 90567820395y^2 + 414148101700y \\ & + 994631470000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right)s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $971142760000 = 2^6 \cdot 5^4 \cdot 7^3 \cdot 70783$:

$$\begin{pmatrix} 8670917500 & 4994192588 & 1482571430 & 1371629620 & 1639388807 & 2152990606 & 6558211230 & 7908670963 & 6997112259 & 1914505825 \\ 0 & 14 & 2 & 12 & 6 & 11 & 8 & 4 & 12 & 6 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 3 & 3 & 2 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 222 factors with signed exponents of absolute value up to 22900082111962766400; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 3325)y^3 + (16s - 43464)y^2 + (-77s + 273390)y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 6645y^8 - 73614y^7 + 24576112y^6 + 699498510y^5 + \\ & 1372949137y^4 - 60244085592y^3 + 90567820395y^2 + 414148101700y \\ & + 994631470000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right)s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $24339292128000 = 2^8 \cdot 3^6 \cdot 5^3 \cdot 1043351$:

$$\begin{pmatrix} 3756063600 & 1306443960 & 2092227624 & 404417550 & 2681935689 & 3266162010 & 1171144293 & 1326581216 & 24929871 & 585551978 \\ 0 & 180 & 135 & 153 & 34 & 49 & 109 & 33 & 34 & 24 \\ 0 & 0 & 3 & 0 & 1 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 12 & 11 & 11 & 11 & 7 & 7 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 205005928259897543418; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2403y^3 + (-40s + 20)y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4806y^8 - 861797y^6 + 4538452582y^4 + 3270477952929y^2 + 730979677478476$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right)s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $123319245609877 = 107 \cdot 1493 \cdot 17579 \cdot 43913$:

$$\begin{pmatrix} 123319245609877 & 95352786100400 & 58353283607816 & 51403374606290 & 67807647851331 & 99771014603640 & 110865057524544 & 48005829327421 & 56508900433645 & 22348932069708 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 593 factors with signed exponents of absolute value up to 145379391786; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2403y^3 + (-40s + 20)y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4806y^8 - 861797y^6 + 4538452582y^4 + 3270477952929y^2 + 730979677478476$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right)s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $152785253526336 = 2^6 \cdot 3^4 \cdot 349 \cdot 84448321$:

$$\begin{pmatrix} 353669568348 & 202585852026 & 297174818703 & 131798455926 & 316398859611 & 248791868016 & 143126282578 & 71985663614 & 132592436906 & 55695810461 \\ 0 & 6 & 0 & 0 & 1 & 2 & 0 & 4 & 3 & 5 \\ 0 & 0 & 3 & 0 & 2 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 6 & 2 & 2 & 1 & 5 & 2 & 3 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 610 factors with signed exponents of absolute value up to 813188462727; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2403y^3 + (-40s + 20)y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4806y^8 - 861797y^6 + 4538452582y^4 + 3270477952929y^2 + 730979677478476$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right)s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7081532492868 = 2^2 \cdot 3^2 \cdot 31^2 \cdot 11743 \cdot 17431$:

$$\begin{pmatrix} 38072755338 & 21737529375 & 17776367461 & 7543254282 & 37315888066 & 15227899647 & 7445854092 & 10117071547 & 2965944340 & 17983903359 \\ 0 & 93 & 44 & 23 & 80 & 59 & 38 & 85 & 40 & 82 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 603 factors with signed exponents of absolute value up to 578035298114; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)y^2 + (18s - 1534266)y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 1712y^8 - 43228y^7 - 2239030y^6 + 26787720y^5 + 7364105347y^4 + 96516078642y^3 - 2914292763072y^2 + 24014800896696y + 2068919484648336$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right)s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4058422565076 = 2^2 \cdot 3^2 \cdot 11^2 \cdot 931685621$:

$$\begin{pmatrix} 61491250986 & 23087792937 & 23920533138 & 20242310435 & 30102312863 & 33156826612 & 58625072829 & 35413614715 & 25177991972 & 49946227477 \\ 0 & 33 & 7 & 6 & 30 & 12 & 12 & 3 & 15 & 17 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 437 factors with signed exponents of absolute value up to 556703612553; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)y^2 + (18s - 1534266)y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 1712y^8 - 43228y^7 - 2239030y^6 + 26787720y^5 + 7364105347y^4 + 96516078642y^3 - 2914292763072y^2 + 24014800896696y + 2068919484648336$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right) s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2132968950288 = 2^4 \cdot 3^2 \cdot 11 \cdot 25939 \cdot 51913$:

$$\begin{pmatrix} 177747412524 & 136217226853 & 91605633517 & 147112646442 & 156275829024 & 36716626989 & 133151893264 & 168662456848 & 77417134960 & 79758755746 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 434 factors with signed exponents of absolute value up to 694224850486; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)y^2 + (18s - 1534266)y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 1712y^8 - 43228y^7 - 2239030y^6 + 26787720y^5 + 7 \\ & 364105347y^4 + 96516078642y^3 - 2914292763072y^2 + 24014800896696 \\ & y + 2068919484648336 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right) s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1309789099076 = 2^2 \cdot 3229 \cdot 101408261$:

$$\begin{pmatrix} 654894549538 & 559463377670 & 34851531669 & 49392909260 & 124295093515 & 479465083738 & 518287054223 & 271375124220 & 563443638587 & 495944555624 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 445 factors with signed exponents of absolute value up to 1006797776379; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2533y^3 + (-12s + 6)y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5066y^8 + 16784799y^6 + 28107615434y^4 + 20701232202625y^2 + 5172655071847500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right)s, \quad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1587485089755 = 3^4 \cdot 5 \cdot 59 \cdot 181 \cdot 367049$:

$$\begin{pmatrix} 58795744065 & 2475000921 & 1251773943 & 9878011400 & 53443592271 & 48938037564 & 36762774567 & 58039699757 & 58019424512 & 36227444774 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 243 factors with signed exponents of absolute value up to 402461500; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2533y^3 + (-12s + 6)y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5066y^8 + 16784799y^6 + 28107615434y^4 + 20701232202625y^2 + 5172655071847500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right)s, \quad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $106972481550400 = 2^6 \cdot 5^2 \cdot 171043 \cdot 390883$:

$$\begin{pmatrix} 1337156019380 & 608178837060 & 255421563030 & 71537238502 & 1197318463745 & 17724436893 & 5868104429 & 246923783624 & 714713308995 & 1053415292066 \\ 0 & 20 & 18 & 13 & 6 & 14 & 10 & 0 & 18 & 5 \\ 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 228 factors with signed exponents of absolute value up to 258157777; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2533y^3 + (-12s + 6)y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5066y^8 + 16784799y^6 + 28107615434y^4 + 20701232202625y^2 + 5172655071847500$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right)s, \quad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $44221102976000 = 2^{10} \cdot 5^3 \cdot 173 \cdot 1996979$:

$$\begin{pmatrix} 69095473400 & 43020677392 & 57928537602 & 63924889306 & 28361252058 & 60811770029 & 40899964314 & 48446504660 & 13439243281 & 42401016237 \\ 0 & 8 & 6 & 4 & 4 & 2 & 4 & 7 & 7 & 3 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 10 & 0 & 4 & 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 223 factors with signed exponents of absolute value up to 162971751; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.